

《流体力学》下半学期总回顾

第五章 理想流体基本方程（对应书第4、5章）

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \vec{V}) = 0$$
$$\frac{D}{Dt} \left(\rho + \frac{V^2}{2} \right) = \vec{f} \cdot \vec{V} + \frac{\nabla \cdot (\vec{P} \cdot \vec{V})}{\rho} + \vec{q}_R$$
$$\text{或 } \vec{q}_R = \frac{D\vec{e}}{Dt} + \rho \frac{D}{Dt} \left(\frac{1}{\rho} \right)$$

$$\frac{D\vec{V}}{Dt} = \vec{f} - \frac{\nabla p}{\rho}$$

Bernoulli方程（定常）柯西-拉格朗日方程（非定常）
沿流线/沿涡线/无旋全场

•无粘，正压，质量力有势

$$\frac{V^2}{2} + \zeta + \Pi = \text{Const}$$
$$\int_1^2 \frac{\partial \vec{V}}{\partial t} \cdot d\vec{l} + \frac{1}{2} V_1^2 + \zeta_1 + \Pi_1 = \frac{1}{2} V_2^2 + \zeta_2 + \Pi_2$$
$$\frac{\partial \varphi}{\partial t} + \frac{V^2}{2} + \zeta + \Pi = c(t)$$

•无粘，不可压，重力

$$\frac{V^2}{2} + \frac{p}{\rho} + gz = \text{Const}$$
$$\int_1^2 \frac{\partial \vec{V}}{\partial t} \cdot d\vec{l} + \frac{p_1}{\rho} + \frac{1}{2} V_1^2 + gz_1 = \frac{p_2}{\rho} + \frac{1}{2} V_2^2 + gz_2$$
$$\frac{\partial \varphi}{\partial t} + \frac{V^2}{2} + \frac{p}{\rho} + gz = c(t)$$

Kelvin定理
Lagrange定理
Helmholtz定理

← 无粘，正压，质量力有势、连续场

二维不可压势流

无旋流动：势函数 φ
 $\vec{\Omega} = \nabla \times \vec{V} = 0$
 $\vec{V} = \nabla \varphi$

(定常/不可压)二维流函数 ψ
自动满足连续方程
基本性质

平面不可压缩流体

$$u = \frac{\partial \varphi}{\partial x}, v = \frac{\partial \varphi}{\partial y}, w = \frac{\partial \varphi}{\partial z}$$
$$V_r = \frac{\partial \varphi}{\partial r}, V_\theta = \frac{1}{r} \frac{\partial \varphi}{\partial \theta}, V_z = \frac{\partial \varphi}{\partial z}$$
$$V_R = \frac{\partial \varphi}{\partial R}, V_\theta = \frac{1}{R} \frac{\partial \varphi}{\partial \theta}, V_\varepsilon = \frac{1}{R \sin \theta} \frac{\partial \varphi}{\partial \varepsilon}$$

$$u = \frac{\partial \psi}{\partial y}, v = -\frac{\partial \psi}{\partial x}$$
$$v_r = \frac{1}{r} \frac{\partial \psi}{\partial \theta}, v_\theta = -\frac{\partial \psi}{\partial r}$$

二维不可压势流

平面不可压无旋流动基本方程：
 $\nabla^2 \varphi = 0, \nabla^2 \psi = 0,$
 φ, ψ 之间关系

平面不可压无旋流动基本解
均匀来流
点源/点汇
点涡

压力--Bernoulli方程的应用

平面不可压势流叠加流动
偶极子
钝体绕流
圆柱绕流
镜像法后叠加

	$\phi = xU_0 \cos \alpha + yU_0 \sin \alpha$	$\psi = yU_0 \cos \alpha - xU_0 \sin \alpha$	$\bar{\phi} = U_0 \cos \alpha - U_0 \sin \alpha$
均匀来流	$\phi = xU_0$	$\psi = yU_0$	$\bar{\phi} = U_0, \bar{\psi} = 0$
平面点源 (Q>0) 平面点汇 (Q<0)	$\phi = \frac{Q}{2\pi} \ln \sqrt{(x-x_0)^2 + (y-y_0)^2}$	$\psi = \frac{Q}{2\pi} \arctg \frac{y-y_0}{x-x_0}$	$\bar{\phi} = \frac{Q(x-x_0)}{2\pi[(x-x_0)^2 + (y-y_0)^2]}$ $\bar{\psi} = -\frac{Q(y-y_0)}{2\pi[(x-x_0)^2 + (y-y_0)^2]}$
平面点涡	$\phi = -\frac{\Gamma}{2\pi} \arctg \frac{y-y_0}{x-x_0}$	$\psi = \frac{\Gamma}{2\pi} \ln \sqrt{(x-x_0)^2 + (y-y_0)^2}$	$\bar{\phi} = -\frac{\Gamma(y-y_0)}{2\pi[(x-x_0)^2 + (y-y_0)^2]}$ $\bar{\psi} = \frac{\Gamma(x-x_0)}{2\pi[(x-x_0)^2 + (y-y_0)^2]}$
x向偶极子	$\phi = \frac{m}{2\pi} \frac{x-x_0}{(x-x_0)^2 + (y-y_0)^2}$	$\psi = -\frac{m}{2\pi} \frac{y-y_0}{(x-x_0)^2 + (y-y_0)^2}$	$\bar{\phi} = -\frac{m \cos \theta}{2\pi r^2}, \bar{\psi} = \frac{m \sin \theta}{2\pi r^2}$
无环量圆柱绕流	$\phi = U_0(r + \frac{a^2}{r}) \cos \theta$	$\psi = U_0(r - \frac{a^2}{r}) \sin \theta$	$\bar{\phi} = U_0(1 + \frac{a^2}{r^2}) \cos \theta, \bar{\psi} = U_0(1 - \frac{a^2}{r^2}) \sin \theta$
有环量圆柱绕流 逆时针Γ>0	$\phi = U_0(r + \frac{a^2}{r}) \cos \theta - \frac{\Gamma}{2\pi} \theta$	$\psi = U_0(r - \frac{a^2}{r}) \sin \theta + \frac{\Gamma}{2\pi} \ln r$	$\bar{\phi} = U_0(1 + \frac{a^2}{r^2}) \cos \theta - U_0(1 - \frac{a^2}{r^2}) \sin \theta - \frac{\Gamma}{2\pi r}$
平面绕排绕流	$\phi = U_0 r \cos \theta + U_0 b \ln r$	$\psi = U_0 r \sin \theta + U_0 b \theta$	$\bar{\phi} = U_0 \cos \theta + \frac{U_0 b}{r}, \bar{\psi} = U_0 \sin \theta$
圆环绕流	$\phi = U_0(R + \frac{a^2}{2R}) \cos \theta$	$\psi = \frac{1}{2} U_0(R^2 - \frac{a^2}{R}) \sin \theta$	$\bar{\phi} = U_0(1 + \frac{a^2}{R^2}) \cos \theta, \bar{\psi} = -U_0(1 - \frac{a^2}{2R^2}) \sin \theta$

第六章 气体动力学 (书第7章)

一维定常绝热等熵流动

$\frac{T}{T_0} = (1 + \frac{\gamma-1}{2} M^2)^{-1}$

也适合绝热间断面

$\frac{T}{T^*} = \frac{\gamma+1}{2 + (\gamma-1)M^2}$

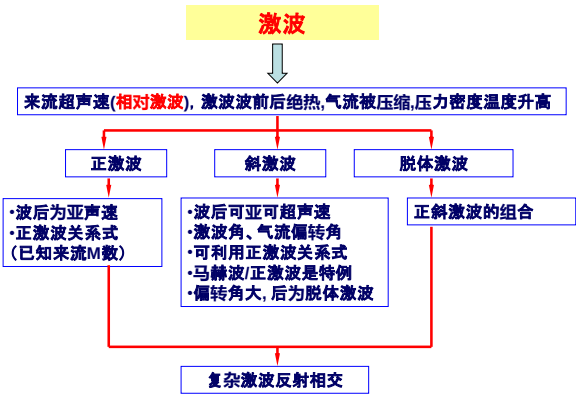
$\frac{p}{p_0} = (1 + \frac{\gamma-1}{2} M^2)^{-\frac{\gamma}{\gamma-1}}$
 $\frac{p}{p_0} = (1 + \frac{\gamma-1}{2} M^2)^{-\frac{1}{\gamma-1}}$

$\frac{p}{p^*} = (\frac{\gamma+1}{2 + (\gamma-1)M^2})^{\frac{\gamma}{\gamma-1}}$
 $\frac{p}{p^*} = (\frac{\gamma+1}{2 + (\gamma-1)M^2})^{\frac{1}{\gamma-1}}$

406页表4

$\frac{A}{A^*} = \frac{1}{M} (\frac{2 + (\gamma-1)M^2}{\gamma+1})^{\frac{\gamma+1}{2(\gamma-1)}} \geq 1$

$Q_m = 0.04042 \frac{p_0}{\sqrt{T_0}} A^* \quad (\gamma = 1.4, R = 287 \text{ J/kg.K})$



式1 $M_2 = \frac{1 + [(\gamma-1)/2] M_1^2}{\gamma M_1^2 - (\gamma-1)/2}$ 正激波 (相对激波) $M_1 > 1, M_2 < 1$

式2 $\frac{\rho_2}{\rho_1} = \frac{V_1}{V_2} = \frac{\lambda_1}{\lambda_2} = \frac{(\gamma+1)M_1^2}{2 + (\gamma-1)M_1^2} > 1$

式3 $\frac{p_2}{p_1} = 1 + \frac{2\gamma}{\gamma+1} (M_1^2 - 1) > 1$

414页表5

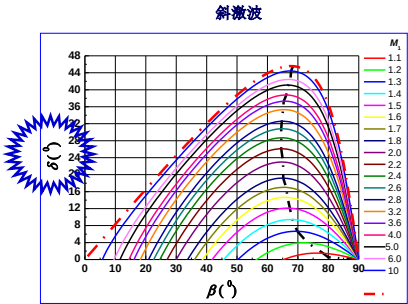
式4 $\frac{T_2}{T_1} = \left[1 + \frac{2\gamma}{\gamma+1} (M_1^2 - 1) \right] \frac{2 + (\gamma-1)M_1^2}{(\gamma+1)M_1^2} > 1$

式5 $\frac{A_1^*}{A_2^*} = \frac{p_{02}}{p_{01}} = \frac{p_1}{p_01} \frac{p_{02}}{p_2} \frac{p_2}{p_1} = \left(\frac{\gamma+1}{2} M_1^2 \right)^{\frac{\gamma}{\gamma-1}} \left(\frac{2\gamma}{\gamma+1} M_1^2 - \frac{\gamma-1}{\gamma+1} \right)^{-\frac{1}{\gamma-1}} < 1$

式6 $\frac{p_{02}}{p_1} = \frac{p_{01}}{p_1} \frac{p_{02}}{p_{01}} = \left(\frac{\gamma+1}{4\gamma M_1^2 - 2(\gamma-1)} \right)^{\frac{\gamma}{\gamma-1}} \frac{1 - \gamma + 2\gamma M_1^2}{\gamma+1}$

$$\operatorname{tg} \delta=\frac{2 \operatorname{ctg} \beta\left(M_1^2 \sin ^2 \beta-1\right)}{(\gamma+1) M_1^2-2\left(M_1^2 \sin ^2 \beta-1\right)}$$

式7



254页图7.15

正激波关系

式1

$$M_2=\left(\frac{1+(\gamma-1) / 2 M_1^2}{\gamma M_1^2-(\gamma-1) / 2}\right)^{\frac{1}{2}}$$

正激波（相对激波）
 $M_1>1, M_2<1$

式2

$$\frac{\rho_2}{\rho_1}=\frac{V_1}{V_2}=\frac{\lambda_1}{\lambda_2}=\frac{(\gamma+1) M_1^2}{2+(\gamma-1) M_1^2}>1$$

式3

$$\frac{p_2}{p_1}=1+\frac{2 \gamma}{\gamma+1}\left(M_1^2-1\right)>1$$

式4

$$\frac{T_2}{T_1}=\left[1+\frac{2 \gamma}{\gamma+1}\left(M_1^2-1\right)\right] \frac{2+(\gamma-1) M_1^2}{(\gamma+1) M_1^2}>1$$

式5

$$\frac{A_1^*}{A_2^*}=\frac{p_{02}}{p_{01}}=\frac{p_1}{p_{01}} \frac{p_{02}}{p_2} \frac{p_2}{p_1}=\left(\frac{\gamma+1}{1+\frac{\gamma-1}{2} M_1^2}\right)^{\frac{\gamma}{\gamma-1}}\left(\frac{2 \gamma}{\gamma+1} M_1^2-\frac{\gamma-1}{\gamma+1}\right)^{\frac{-1}{\gamma-1}}<1$$

将正激波关系中
 M_1 用 $M_{1n}=M_1 \sin \beta$ 代替
 M_2 用 $M_{2n}=M_2 \sin (\beta-\delta)$ 代替



小扰动的膨胀和压缩波

因为是小扰动, 整个过程是等熵的

419页表6

Prandtl-Meyer角 θ

$$\theta=\sqrt{\frac{\gamma+1}{\gamma-1}} \arctg \sqrt{\frac{\gamma-1}{\gamma+1}\left(M^2-1\right)}-\arctg \sqrt{M^2-1}$$

θ 为气流从 $M=1, \theta=0$ 到 M 流动的偏转角

从 M_1 到 M_2 膨胀加速偏转气流, $\theta_2\left(M_2\right) > \theta_1\left(M_1\right)$
气流角度偏转了 $\delta=\theta_2-\theta_1$ (相对 M_1 来流方向)

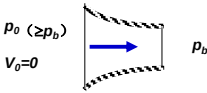
从 M_1 到 M_2 压缩减速偏转气流, $\theta_2\left(M_2\right) < \theta_1\left(M_1\right)$
气流角度偏转了 $\delta=\theta_1-\theta_2$ (相对 M_1 来流方向)

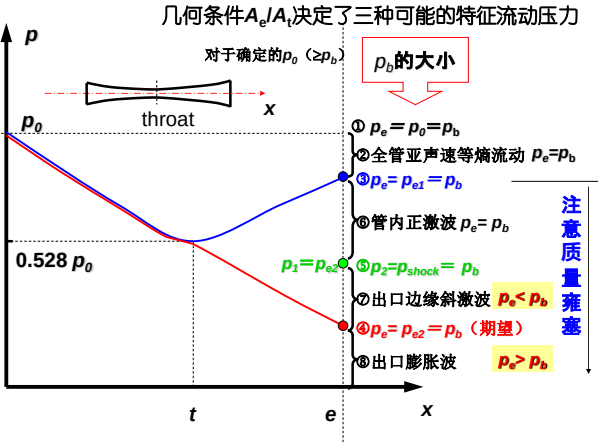
渐缩喷管中的等熵流动, 环境压力 p_b

1. $p_b > p^*$ 出口处不会达到声速流动, 全管亚声速, 且 $p_e=p_b$

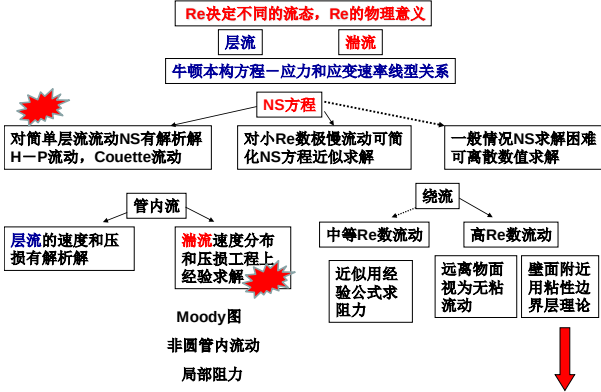
2. $p_b \leq p^*$ 出口处(最小截面)达到声速流动, $p_e=p^*$, 管内亚声速,

质量堵塞





第七章 粘性流体力学 (书第8、9、10章)



边界层理论

高Re数无(小)分离绕流问题

边界层概念及力学特征

外场的势流的求解

Euler方程

边界层内流动的求解

Prandtl 边界层偏微分方程

Biacius相似解

Von Karman动量积分方程

假设速度后的近似解

求边界层各种厚度和摩擦阻力

平板边界层求解实例

全板层流边界层

全板湍流边界层

板上层湍流混合边界层